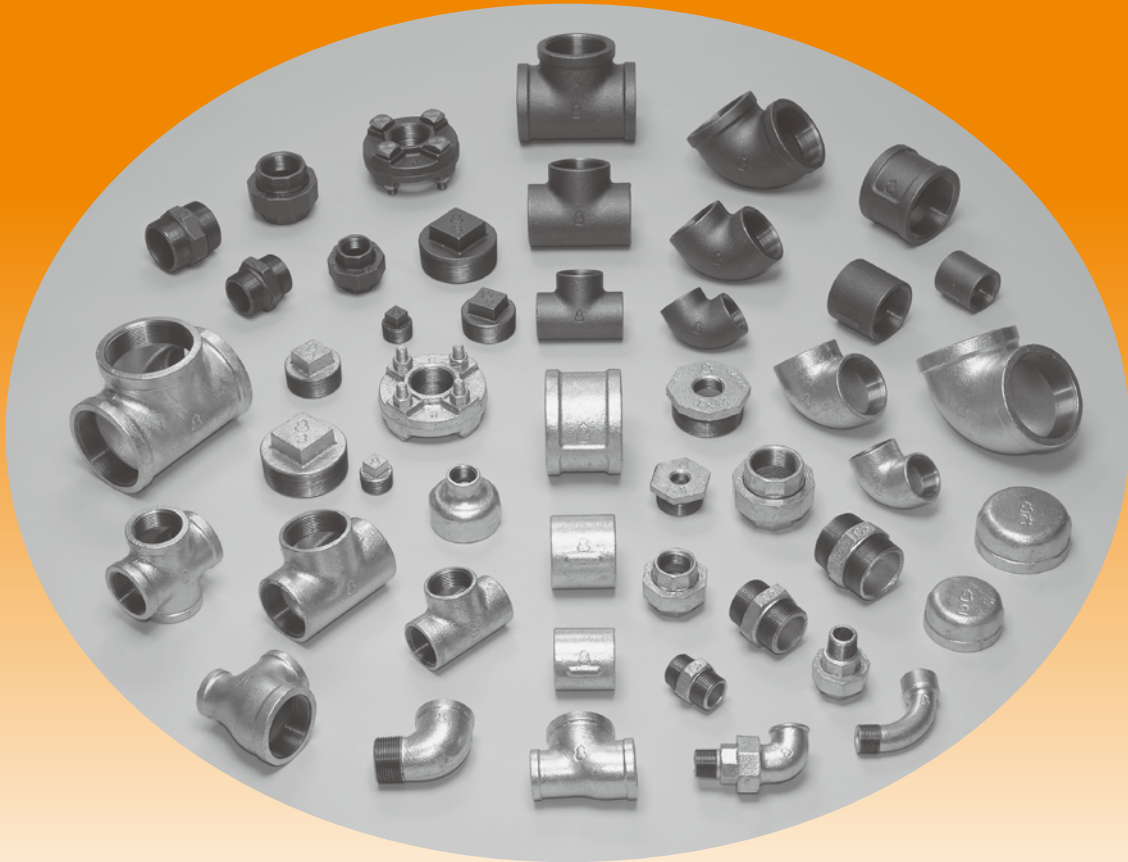


GOURD  BRAND

MALLEABLE IRON PIPE FITTINGS

MALLEABLE IRON PIPE FITTINGS



Gourd  Brand malleable iron pipe fittings are manufactured by Hitachi Metals, Ltd., the leading manufacturer of malleable iron pipe fittings in the world with a century of experience.

The high quality of Gourd  Brand pipe fittings is achieved by strict quality control in every stage of the manufacturing process.

Gourd  Brand has lived up to its reputation for excellence in quality, variety, and punctual delivery, which is known the world over.

Contents

Technical Information	3
Ordering Information	3
Available Items and Sizes, Standard Class 150	4
Outline for Standard Class 300	11

Technical Information

1) Material

Gourd Brand pipe fittings made of black heart malleable cast iron to ASTM A197

2) Spec

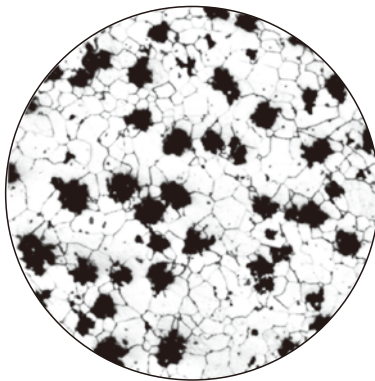
ANSI/ASME B16.3, B16.14, B16.39

3) Thread

ANSI/ASME B1.20.1(NPT), ISO7-1:1994(BS/DIN), BS21, DIN2999

4) Galvanized

ANSI/ASTM A153 / A153M - 09, BS EN ISO 1461: 2009



Microstructure of the Gourd Brand Black Heart Malleable Iron after Annealing (x100)

5) Pressure - Temperature Ratings

Maximum Working Pressure for Class 150lbs	
Temperature(°C)	Bar
-29 to 66	20.7
100	17.5
125	15.2
150	12.8
175	10.5*1

*1 Permissible for service temperature up to 186°C, reflecting the temperature of saturated steam at 10.3 bar.

General Notes;

(a) 1bar = 14.5psi = 100kPa

(b) $^{\circ}\text{C} = \frac{^{\circ}\text{F}-32}{1.8}$

6) Tolerances

Size	Tolerance(mm)	Size	Tolerance(mm)
1/8	±0.8	1 1/2,2	±2.0
1/4	±1.0	2 1/2,3	±2.5
3/8	±1.3	4,5	±3.0
1/2,3/4	±1.5	6	±3.6
1, 1 1/4	±1.8		

Ordering Information

1) When ordering the Gourd Brand Pipe Fittings, please specify the requirements:

- Type of thread.
- Black or Galvanized.
- Fig. No.
- Name, size and quantity.

2) To minimize handling costs, Gourd Brand Pipe Fittings shall require quantities suitable for exports.

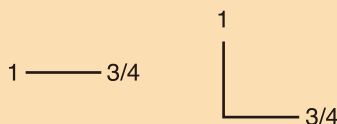
3) To avoid confusion, Gourd Brand Pipe Fittings with different diameters should be described as follows:

A Two Ways Fittings

Larger diameter comes first, and smaller second.

(Example)

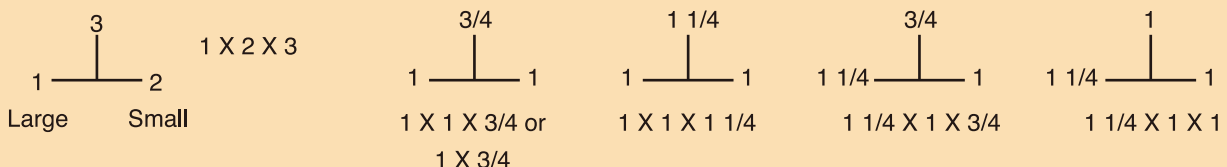
1 X 3/4



B Three Ways Fittings

Larger diameter of the two on the same line comes first and smaller second, and the remaining third.

(Example)

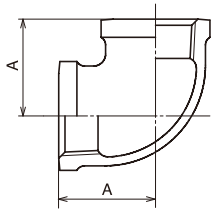


Available Items and Sizes, Standard Class 150.

Elbows 90° Banded Equal

Fig.002

BL

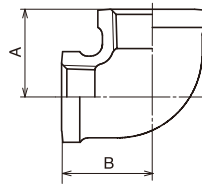


Size (Inch)	Dimension A
1/8	17.5
1/4	20.6
3/8	24.1
1/2	28.5
3/4	33.3
1	38.1
1 1/4	44.5
1 1/2	49.3
2	57.2
2 1/2	68.6
3	78.2
4	96.3
5	114.3
6	130.3

Elbows 90° Banded Reducing

Fig.002

BRL

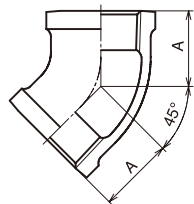


Size (Inch)	Dimensions	
	A	B
1/4X1/8	18.8	19.3
3/8X1/8	20.5	21.6
3/8X1/4	22.4	22.9
1/2X1/4	24.6	24.9
1/2X3/8	26.4	26.2
3/4X1/2	30.5	31.0
1X1/2	32.0	35.5
1X3/4	34.8	36.8
1 1/4X1/2	34.0	35.9
1 1/4X1	40.1	42.4
1 1/2X1/2 *	37.9	45.2
1 1/2X3/4	38.6	44.5
1 1/2X1	41.9	45.7
1 1/2X1 1/4	46.0	48.0
2X3/4	40.6	50.0
2X1	43.9	51.3
2X1 1/4	48.0	53.0
2X1 1/2	52.0	55.0
2 1/2X2	61.0	66.0
3X1 *	57.6	69.8
3X2	64.0	73.4

Elbows 45° Banded Equal

Fig.004

BL45

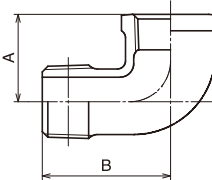


Size (Inch)	Dimensions A
1/2	22.4
3/4	24.9
1	28.5
1 1/4	32.8
1 1/2	36.3
2	42.7
2 1/2	49.5
3	55.1
4	66.3
6	87.9

Street Elbows 90° Banded Equal

Fig.006

SL

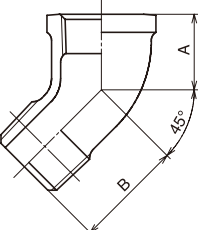


Size (Inch)	Dimensions	
	A	B
1/8	17.5	25.4
1/4	20.6	30.2
3/8	24.1	36.6
1/2	28.5	41.2
3/4	33.3	48.0
1	38.1	54.4
1 1/4	44.5	62.2
1 1/2	49.3	68.3
2	57.2	82.8
2 1/2	68.6	98.0
3	78.1	114.6
4	96.3	144.5

Street Elbows 45° Banded Equal

Fig.007

SL45

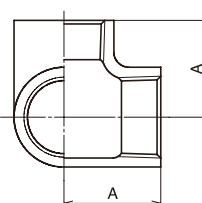


Size (Inch)	Dimensions	
	A	B
1/2	22.4	29.2
3/4	24.9	32.8
1	28.5	37.3
1 1/4	32.8	43.4
1 1/2	36.3	47.8
2	42.7	65.3

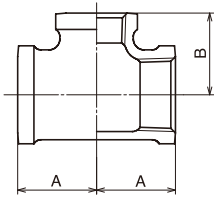
Side Outlet Elbows Banded *

Fig.009

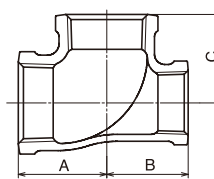
SOL



Size (Inch)	Dimensions A
1/2	29.3
3/4	34.1
1	39.3
1 1/4	45.3
1 1/2	50.6
2	58.5

Tees Banded Reducing
Fig.011
BRT


Size (Inch)	Dimensions	
	A	B
3/8X1/4	22.4	22.9
1/2X1/8 *	22.6	23.5
1/2X1/4	24.6	24.9
1/2X3/8	26.4	26.2
3/4X3/8	28.5	28.7
3/4X1/2	30.5	31.0
1X1/4	28.2	31.0
1X3/8	30.0	32.3
1X1/2	32.0	35.5
1X3/4	34.8	36.8
1 1/4X1/2	34.0	38.9
1 1/4X3/4	36.8	41.2
1 1/4X1	40.1	42.4
1 1/2X1/2	35.8	42.2
1 1/2X3/4	38.6	44.5
1 1/2X1	41.9	45.7
1 1/2X1 1/4	46.2	47.8
2X1/2	37.9	47.8
2X3/4	40.6	50.0
2X1	43.9	51.3
2X1 1/4	48.3	55.3
2X1 1/2	51.3	54.9
2 1/2X1/2 *	42.5	57.9
2 1/2X3/4	44.2	58.9
2 1/2X1	47.5	60.2
2 1/2X1 1/4	51.8	62.2
2 1/2X1 1/2	54.9	62.8
2 1/2X2	60.7	66.0
3X1/2	44.2	66.5
3X3/4	47.5	66.3
3X1	50.8	67.6
3X1 1/4	55.1	69.6
3X1 1/2	58.2	71.1
3X2	64.6	73.4
3X2 1/2	71.9	76.0
4X1 *	58.4	83.8
4X1 1/4*	62.3	87.4
4X1 1/2	63.4	84.3
4X2	69.6	86.6
4X2 1/2	77.5	88.2
4X3	83.8	91.4
6X4	104.9	125.5

Tees Banded Reducing
Fig.011
BRT


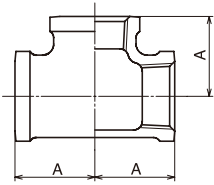
Size (Inch)	Dimensions		
	A	B	C
1/2X1/2X3/4 *	30.8	30.8	31.1
1/2X1/2X1 *	33.7	33.7	33.2
3/4X1/2X1/2	30.5	28.5	31.0
3/4X1/2X3/4	32.3	31.0	33.2
3/4X3/4X1 *	35.7	35.7	35.3
3/4X3/4X1 1/4	41.2	41.2	36.8
3/4X3/4X1 1/2 *	-	-	-
1X1/2X1/2	32.0	38.5	35.5
1X1/2X3/4	34.8	31.0	36.8
1X1/2X1	38.1	35.5	38.1
1X3/4X1/2	32.0	30.5	35.5
1X3/4X3/4	34.8	33.3	36.8
1X3/4X1	38.1	36.8	38.1
1X1X1 1/4	42.4	42.4	40.1
1X1X1 1/2	45.7	45.7	41.9
1 1/4X1/2X3/4 *	37.8	31.1	41.4
1 1/4X1/2X1	40.1	34.5	42.4
1 1/4X1/2X1 1/4	44.5	38.9	44.5
1 1/4X3/4X1 1/4	44.5	41.2	44.5
1 1/4X1X1/2	34.0	32.0	38.9
1 1/4X1X3/4	36.8	34.8	41.2
1 1/4X1X1	40.1	38.1	42.4
1 1/4X1X1 1/4	44.5	42.4	44.5
1 1/4X1 1/4X1 1/2	47.8	47.8	46.2
1 1/2X1/2X1 1/2	49.3	42.2	49.3
1 1/2X1X1	41.9	38.1	45.7
1 1/2X1X1 1/4	46.2	42.4	47.8
1 1/2X1X1 1/2	49.3	45.7	49.3
1 1/2X1 1/4X1/2	35.8	34.0	42.2
1 1/2X1 1/4X3/4	38.6	36.3	44.5
1 1/2X1 1/4X1	41.9	40.1	45.7
1 1/2X1 1/4X1 1/4	46.2	44.5	47.8
1 1/2X1 1/4X1 1/2	49.3	47.8	49.3
1 1/2X1 1/2X2	54.9	54.9	51.3
2X1/2X2	57.2	47.8	57.2
2X3/4X2	57.2	50.0	57.2
2X1X2	57.2	51.3	57.2
2X1 1/4X1 *	45.2	42.0	53.1
2X1 1/4X1 1/4	48.3	44.5	53.3
2X1 1/2X3/4 *	42.3	38.9	51.7
2X1 1/2X1	43.9	41.9	51.3
2X1 1/2X1 1/4	48.3	46.2	53.3
2X1 1/2X1 1/2	51.3	49.3	54.9
2 1/2X2X1 1/2	56.4	53.4	63.0
2 1/2X2X2	60.7	57.2	66.0
3X2X2	65.5	57.9	74.6

Available Items and Sizes, Standard Class 150.

Tees Banded Equal

Fig.011

BT

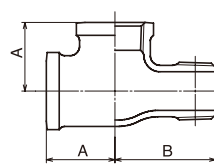


Size (Inch)	Dimension A
1/8	17.5
1/4	20.6
3/8	24.1
1/2	28.5
3/4	33.3
1	38.1
1 1/4	44.5
1 1/2	49.3
2	57.2
2 1/2	68.6
3	78.2
4	96.3
5	114.3
6	130.3

Service Tees Banded Equal

Fig.012

ST

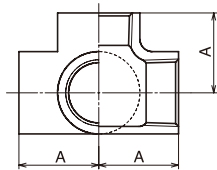


Size (Inch)	Dimension	
	A	B
1/2	28.5	41.4
3/4	33.3	48.0
1	38.1	54.4
1 1/4	44.5	62.2
1 1/2	49.3	68.3
2	57.2	82.8

Side Outlet Tees Plain *

Fig.013

SOT

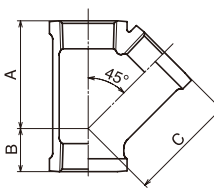


Size (Inch)	Dimension A
1/2	27.7
3/4	32.8
1	39.0
1 1/4	47.4
1 1/2	49.4
2	58.4

Y Baranches 45° Banded

Fig.014

BY45

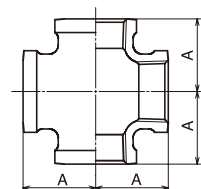


Size (Inch)	Dimension		
	A	B	C
1/2	45.4	15.5	43.4
3/4	52.1	18.5	52.1
1	61.7	21.6	61.7
1 1/4	74.2	25.9	74.2
1 1/2	83.5	29.9	83.5
2	99.8	37.1	99.8

Cross Banded Equal

Fig.016

BCR

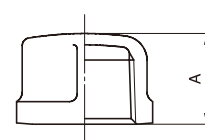


Size (Inch)	Dimension A
1/2	28.5
3/4	33.3
1	38.1
1 1/4	44.5
1 1/2	49.3
2	57.2
2 1/2	68.6
3	78.2
4	96.3

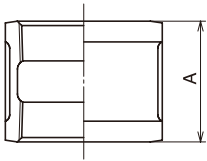
Caps Banded

Fig.018

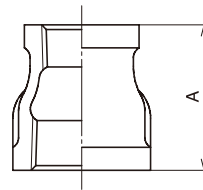
BCA



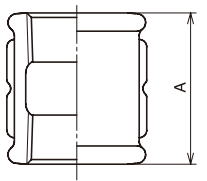
Size (Inch)	Dimension A
1/8	13.5
1/4	16.0
3/8	18.8
1/2	22.1
3/4	24.6
1	29.5
1 1/4	32.5
1 1/2	33.8
2	36.8
2 1/2	43.2
3	45.7
4	52.8

Sockets Banded Equal With Ribs
Fig.020
BS


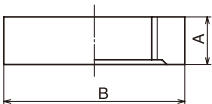
Size (Inch)	Dimension A
1/8	24.2
1/4	26.9
3/8	29.5
1/2	34.0
3/4	38.6
1	42.4
1 1/4	49.0
1 1/2	54.6
2	64.3
2 1/2	73.2
3	80.8
4	93.7
5 *	97.0
6 *	107.3

Reducing Sockets Banded
Fig.022
BRS


Size (Inch)	Dimension A
1/4X1/8	25.4
3/8X1/8	28.7
3/8X1/4	28.7
1/2X1/8	31.8
1/2X1/4	31.8
1/2X3/8	31.8
3/4X3/8	38.6
3/4X1/2	36.6
1X3/8	42.9
1X1/2	42.9
1X3/4	42.9
1 1/4X1/2	52.3
1 1/4X3/4	52.3
1 1/4X1	52.3
1 1/2X1/2	58.7
1 1/2X3/4	58.7
1 1/2X1	58.7
1 1/2X1 1/4	58.7
2X1/2	71.4
2X3/4	71.4
2X1	71.4
2X1 1/4	71.4
2X1 1/2	71.4
2 1/2X1	82.6
2 1/2X1 1/4	82.6
2 1/2X1 1/2	82.6
2 1/2X2	82.6
3X3/4	93.7
3X1	93.7
3X1 1/4	93.7
3X1 1/2	93.7
3X2	93.7
3X2 1/2	93.7
4X2	111.3
4X2 1/2	111.3
4X3	111.3
6X4 *	107.8

Sockets Right & Left Hand Threads With *
Fig.020A
EBS


Size (Inch)	Dimension A
3/8	31.2
1/2	36.0
3/4	40.9
1	46.4
1 1/4	51.2
1 1/2	56.2
2	60.9

Locknuts
Fig.024
LN


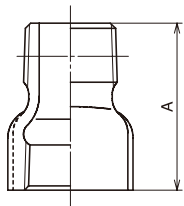
Size (Inch)	Dimension	
	A	B
1/2	7.9	30.0
3/4	8.6	36.3
1	9.7	44.5
1 1/4	10.7	53.3
1 1/2	11.9	59.7
2	13.5	73.2
2 1/2	15.0	88.9
3	17.0	108.5
4	20.3	136.7

Available Items and Sizes, Standard Class 150.

Extension Pieces Banded *

Fig.025

X

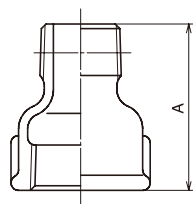


Size (Inch)	Dimensions A
3/8	35.2
1/2	43.2
3/4	49.7
1	56.8
1 1/4	62.3
1 1/2	66.9
2	72.3

Sokets Reducing Male & Female *

Fig.025A

MFS

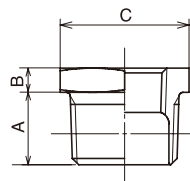


Size (Inch)	Dimensions A
1/2X3/8	43.6
3/4X1/2	49.3
1X3/4	56.5
1 1/4X1/2	59.4
1 1/4X1	61.5
1 1/2X3/4	61.6
1 1/2X1	61.8
1 1/2X1 1/4	64.6
2X1	69.9
2X1 1/4	70.1
2X1 1/2	71.6

Bushings

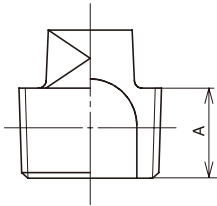
Fig.026

BU

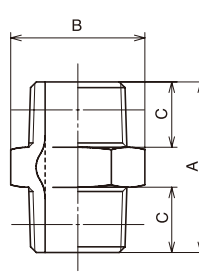


Size (Inch)	Dimensions		
	A	B	C
1/4X1/8	11.2	3.6	16.3
3/8X1/8	12.2	4.1	17.3
3/8X1/4	12.2	4.1	17.3
1/2X1/8	14.2	4.8	22.1
1/2X1/4	14.2	4.8	22.1
1/2X3/8	14.2	4.8	22.1
3/4X1/8	16.0	5.6	29.2
3/4X1/4	16.0	5.6	29.2
3/4X3/8	16.0	5.6	29.2
3/4X1/2	16.0	5.6	29.2
1X1/8 *	20.2	7.5	39.0
1X1/4 *	20.2	7.5	39.0
1X3/8 *	22.5	7.7	39.0
1X1/2	19.1	6.4	36.1
1X3/4	19.1	6.4	36.1
1 1/4X1/2 *	22.1	9.1	47.5
1 1/4X3/4	20.3	7.1	44.7
1 1/4X1	20.3	7.1	44.7
1 1/2X3/8 *	25.3	10.0	55.8
1 1/2X1/2 *	25.5	10.0	55.8
1 1/2X3/4 *	25.7	10.0	55.8
1 1/2X1	21.1	7.9	50.8
1 1/2X1 1/4	21.1	7.9	50.8
2X3/8 *	27.6	11.8	65.0
2X1/2 *	27.2	12.2	65.2
2X3/4 *	27.3	11.9	64.8
2X1 *	27.3	11.7	64.3
2X1 1/4	22.4	8.6	63.0
2X1 1/2	22.4	8.6	63.0
2 1/2X3/4 *	31.1	11.6	82.1
2 1/2X1 *	31.1	11.6	82.6
2 1/2X1 1/4 *	30.9	11.0	82.1
2 1/2X1 1/2	27.2	11.2	68.1
2 1/2X2	27.2	9.4	75.7
3X1 *	35.0	12.7	98.0
3X1 1/4 *	35.0	12.7	98.0
3X1 1/2 *	35.0	12.7	98.0
3X2	28.7	12.2	83.3
3X2 1/2	28.7	10.2	98.0
4X1 *	39.7	15.1	123.0
4X1 1/4 *	40.0	14.8	123.3
4X1 1/2 *	40.0	15.4	123.3
4X2 *	40.0	14.9	123.0
4X2 1/2 *	40.6	14.5	123.3
4X3	31.0	12.7	117.3
5X2 *	58.6	15.1	149.0
5X3 *	58.6	15.9	147.5
5X4	33.3	12.7	147.1
6X3 *	66.4	20.0	177.5
6X4 *	66.4	18.6	178.3
6X5	35.6	16.0	179.1

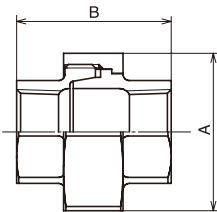


Plugs Plain
Fig.028
P


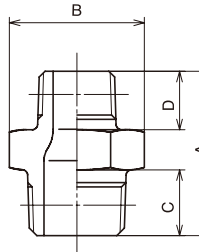
Size (Inch)	Dimensions A
1/4	11.2
3/8	12.2
1/2	14.2
3/4	16.0
1	19.1
1 1/4	20.3
1 1/2	21.1
2	22.4
2 1/2	27.2
3	28.7
4 *	31.0

Nipples Equal *
Fig.032
NI


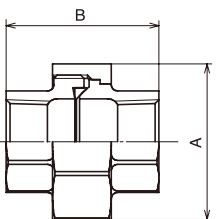
Size (Inch)	Dimension		
	A	B	C
1/8	32.0	14.0	11.0
1/4	34.0	17.0	12.0
3/8	36.0	21.0	13.0
1/2	45.0	26.0	18.5
3/4	48.0	31.0	19.5
1	53.0	38.0	21.5
1 1/4	56.0	46.0	23.0
1 1/2	60.0	54.0	24.0
2	70.0	69.0	29.0
2 1/2	76.0	80.0	29.5
3	84.0	95.0	33.5
4	95.0	120.0	38.5
5	110.0	145.0	44.0
6	116.0	170.0	46.0

Unions Taper Seat Iron to Iron Female *
Fig.031A
UC


Size (Inch)	Dimension	
	A	B
1/4	31.0	40.0
3/8	37.0	43.0
1/2	42.0	46.5
3/4	49.0	50.5
1	59.0	57.0
1 1/4	69.0	64.0
1 1/2	78.0	69.5
2	93.0	76.5
2 1/2	112.0	85.0
3	127.0	94.0
4	158.0	113.0

Nipples Reducing *
Fig.032A
RNI


Size (Inch)	Dimension			
	A	B	C	D
3/8X1/4	38.0	21.0	14.5	13.5
1/2X1/4	42.5	26.0	18.5	14.0
1/2X3/8	43.0	26.0	18.5	14.5
3/4X3/8	43.2	32.3	18.7	13.5
3/4X1/2	47.0	32.0	19.0	17.0
1X1/2	52.0	38.0	22.0	18.0
1X3/4	53.0	38.0	22.0	19.0
1 1/4X3/4	56.0	46.0	23.5	20.5
1 1/4X1/2	52.0	46.0	22.0	18.0
1 1/4X1	59.0	46.0	23.0	24.0
1 1/2X1/2	63.0	54.0	23.0	20.0
1 1/2X1	60.0	54.0	23.0	24.0
1 1/2X1 1/4	68.0	54.0	23.0	23.0
2X1	68.0	63.0	27.5	25.0
2X1 1/4	70.5	63.0	27.5	25.0
2X1 1/2	70.0	63.0	27.5	27.0
2 1/2X1 1/2	68.0	80.0	28.0	23.0
2 1/2X2	70.0	80.0	28.0	25.0
3X2	86.0	95.0	40.0	29.0
3X2 1/2	77.0	95.0	32.0	28.0
4X2	80.0	120.0	37.0	25.0
4X3	87.0	120.0	37.0	32.0

Unions Brass to Iron Seat Female
Fig.031B
US


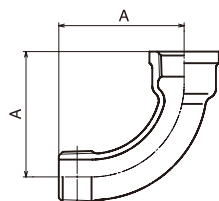
Size (Inch)	Dimension	
	A(min.)	B(min.)
1/2	37.0	43.5
3/4	43.5	49.5
1	52.5	52.5
1 1/4	63.5	57.5
1 1/2	71.5	61.0
2	86.5	70.0
2 1/2	104.5	82.0
3	120.5	89.0
4	152.5	98.0

Available Items and Sizes, Standard Class 150.

Bends 90° Bended Male & Female *

Fig.033B

BMBE

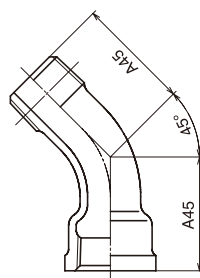


Size (Inch)	Dimensions A
3/8	45.3
1/2	51.8
3/4	65.8
1	82.6
1 1/4	101.2
1 1/2	115.9
2	141.5
2 1/2	173.0
3	206.1
4	260.0

Bends 45° Bended Male & Female *

Fig.033C

BMBE45

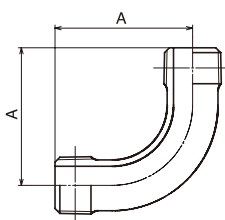


Size (Inch)	Dimensions A
1/2	38.7
3/4	45.4
1	55.8
1 1/4	63.5
1 1/2	70.4
2	86.3
2 1/2	100.0
3	115.0
4	145.0

Bends 90° Male *

Fig.035

OBE

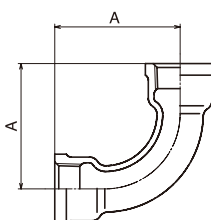


Size (Inch)	Dimension A
1/2	52.4
3/4	66.1

Bends 90° Bended Female *

Fig.036B

BIBE

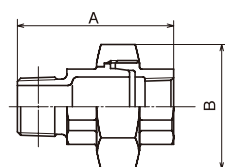


Size (Inch)	Dimension A
3/8	44.0
1/2	52.0
3/4	65.0
1	82.6
1 1/4	100.7
1 1/2	115.9
2	140.0
2 1/2	171.5
3	205.0
4	260.0

Unions Taper Seat Male Female *

Fig.040A

MUC

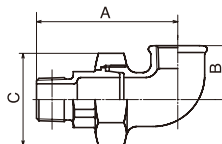


Size (Inch)	Dimension	
	A	B
3/8	57.5	37.0
1/2	65.0	42.0
3/4	71.5	49.0
1	80.8	59.0
1 1/4	90.8	69.0
1 1/2	96.5	78.0
2	107.5	93.0
2 1/2	119.8	112.0
3	131.7	127.0

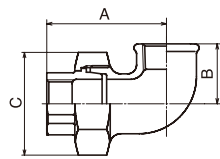
Union Elbows Taper Seat Male & Female *

Fig.043A

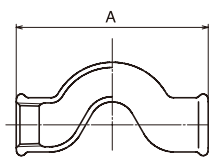
MULC



Size (Inch)	Dimension		
	A	B	C
1/2	78.5	27.0	42.0
3/4	87.0	33.0	49.0
1	98.3	38.0	59.0
1 1/4	110.3	45.0	69.0
1 1/2	122.0	50.0	78.0
2	136.5	58.0	93.0

Unions Elbow Taper Seat Female *
Fig.040A
ULC


Size (Inch)	Dimension		
	A	B	C
1/2	60.5	27.0	42.0
3/4	66.5	33.0	49.0
1	74.8	38.0	59.0
1 1/4	83.8	45.0	69.0
1 1/2	95.5	50.0	78.0
2	106.0	58.0	93.0

Crossovers, Equal *
Fig.050
NOS


Size (Inch)	Dimension A
1/2	93.2
3/4	112.8
1	141.6

*Item based on our own specification.

Outline for Standard Class 300.

Pressure - Temperature Ratings

Maximum Working Pressure for Class 300lbs, Bar				
Temperature(°C)	1/4 - 1	1 1/4 - 2	2 1/2 - 3	Unions
-29 to 66	137.9	103.4	69.0	41.4
100	119.6	90.5	61.5	37.5
125	106.4	91.1	56.1	34.6
150	93.1	71.8	50.7	31.7
175	79.9	62.5	45.2	28.9
200	66.6	53.1	39.8	26.0
225	53.4	43.8	34.3	23.1
250	40.1	34.5	28.9	20.3
275	26.9	25.2	23.4	17.4
288	20.7	20.7	20.7	15.9

Available Items

Fig.	Name and abbreviation		Size (Inch)
650	90° Elbows	HBL	1/2 - 5
652	45° Elbows	HBL45	1/2 - 5
653	90° Street Elbows	HSL	1/2 - 3
654	Tees	HBT	1/2 - 4
655	Reducing Tees	HBRT	3/4X1/2-2X1 1/2
656	Sockets	HBS	1/2 - 3
657	Reducing Sockets	HBRS	3/4X1/2-2X1 1/2
658	Caps	HBCA	1/2 - 3
659	Bushing	HBU	3/4X1/2-4X3
661	Union Brass to Iron Seat Female	HUS	1/2 - 3

For further details, please contact us.



Hitachi Metals, Ltd.

• **Head Office**

SEAVANS North Bldg., 2-1 Shibaura 1-chome, Minato-ku, Tokyo
105-8614, Japan
Tel. +81-3-5765-4280 Fax. +81-3-5765-8313
<http://www.hitachi-metals.co.jp>

Hitachi Metals Europe GmbH

• **Head Office**

Immermannstrasse 14-16, 40210 Dueseldorf, Germany
Tel. +49-211-16009-0 Fax. +49-211-16009-29

• **South Germany Branch**

Romerstrasse 33, 71229 Leonberg, Germany
Tel. +49-7152-93975-0 Fax. +49-7152-93975-20

• **Milano Office**

Milano S-Felice Torre 7, Segrate 20090, Milano, Italy
Tel. +39-02-7530188/7532613/7533782 Fax. +39-02-7532558

• **London Office**

Tudor House, Kingsway Business Park, Oldfield Road, Hampton,
Middlesex, TW12 2HD England
Tel. +44-20-8941-0123 Fax. +44-20-8941-7231

• **Paris Office**

Immeuble "Ariane", 18 rue Grange Dame Rose BP 134, 78148
VELIZY Cedex, France
Tel. +33-1-34.65.00.56 Fax. +33-1-39.46.18.50

Hitachi Metals Singapore Pte. Ltd.

12 Gul Avenue, Singapore 629656
Tel. +65-6861-7711 Fax. +65-6861-9554

Hitachi Metals Hong Kong Ltd.

Units 2212-14, 22/F., Miramar Tower, 132 Nathan Road, Tsimshatsui,
Kowloon, Hong Kong
Tel. +852-2724-4183 Fax. +852-2311-2095

Hitachi Metals (Shanghai) Ltd.

11F, Chong Hing Finance Center, No.288, Nan Jing Road (W),
Shanghai, 200003, China
Tel. +86-21-3366-3000 Fax. +86-21-3366-3030

Hitachi Metals (Suzhou) Technology, Ltd.

220 Xing Ming Street, Suzhou Industrial Park, Suzhou,
Jiangsu, 215021, China
Tel. +86-512-6256-7880 Fax. +86-512-6256-8980

Hitachi Metals America, Ltd.

<http://www.hitachimetals.com>

• **Head quarters in North America**

2 Manhattanville Road, Suite 301, Purchase, NY 10577-2103 U.S.A.
Tel. +1-914-694-9200 Fax. +1-914-694-9279

• **New York Office**

2 Manhattanville Road, Suite 301, Purchase, NY 10577-2103 U.S.A.
Tel. +1-914-694-9200 Fax. +1-914-694-9279

• **San Jose Office**

2107 North First Street, Suite 500, San Jose, CA 95131, U.S.A.
Tel. +1-408-467-8900 Fax. +1-408-467-8901

• **Chicago Office**

2101 South Arlington Heights Road, Suite 116, Arlington Heights,
IL 60005-4142, U.S.A.
Tel. +1-847-364-7200 Fax. +1-847-364-7279

• **Detroit Office**

Pinnacle Office Center, 41800 West Eleven Mile Road, Suite 100,
Novi, MI 48375, U.S.A.
Tel. +1-248-465-6400 Fax. +1-248-465-6020/6021

• **Pittsburgh Office**

222 North Park Drive, Suite 2, West Hills Industrial Park, Kittanning,
PA 16201-8883, U.S.A.
Tel. +1-724-545-3161 Fax. +1-724-545-3164

• **Charlotte Office**

Providence Park, 10700 Sikes Place, Suite 240, Charlotte, NC 28277,
U.S.A.
Tel. +1-704-321-1350 Fax. +1-704-321-1355

Hitachi Metals (Thailand) Ltd.

1/60, Moo 5, Rojana Industrial Park, Tambol Khanharm, Amphur
Uthai, Ayutthaya 13210, Thailand
Tel. +66-35-330-588 Fax. +66-35-330-395

Ward Manufacturing LLC.

P.O.Box 9, 115 Gulick Street, Blossburg, PA 16912-0009, U.S.A.
Tel. +1-570-638-2131 Fax. +1-570-638-3410

All information and data on this brochure are subject to change without notice.

Our address and your contact indicated in this catalog are those as of July 2011.

If you cannot put a call through, please contact Corporate Communication Group in Tokyo below.

Tel:+81-3-5765-4076 Fax:+81-3-5765-8312

E-mail : hmcc@hitachi-metals.co.jp

Catalog No.HE-A7-J

Printed in 2011.7 (M-F T₃)